

A Machine Learning and SUMO-Based Framework for CO₂ Emission Prediction in Urban Areas with Web Application Deployment

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Abstract. The accelerated growth of vehicle fleets in Latin American cities, coupled with high altitudes and heavy traffic congestion, has substantially increased the environmental impact of carbon dioxide (CO_2) emissions. This work presents a practical methodology to predict CO_2 emissions in urban areas, avoiding the need for computationally expensive traffic simulations. To achieve this, a dataset was generated by performing extensive microscopic traffic simulations with SUMO, using [OpenStreetMap data](#) to extract the urban road network and configuring vehicle flows based on official registration statistics from [Ecuador](#). Multiple scenarios with varying vehicle densities and fleet compositions were simulated to build a diverse dataset. Three machine learning models, [Linear Regression, Random Forest, and Neural Networks](#), were trained on this data set to predict CO_2 emissions as a function of input traffic parameters. The Random Forest model outperformed the others, achieving $R^2 = 0.9875$ and $MAPE = 3.61\%$. This trained model was then deployed in a web application using [Streamlit](#), allowing users to estimate emissions in real time by inputting simple traffic parameters, thereby eliminating the need for running new extensive SUMO simulations for each scenario. This framework offers an efficient decision support tool for urban planning and environmental assessment in high-altitude, traffic-congested cities like Quito.

Keywords: CO_2 emissions · SUMO · Machine learning · Web application · OpenStreetMap.

1 Introduction

Urban expansion and the increase in the number of vehicles in high-altitude cities such as Quito have led to an increase in CO_2 emissions contributing to global

warming and thus to climate change [1]. Quito, being located at 2,850 meters above sea level, faces particular conditions because the lower oxygen concentration forces engines to consume more fuel, which is one of the causes of increased CO_2 emissions. [2]. This issue is further compounded by severe traffic congestion, making Quito, according to INRIX, the most congested city in Ecuador and the fourth most congested in South America [3]. Addressing these environmental concerns requires advanced tools capable of replicating real-world traffic scenarios. SUMO (Simulation of Urban MObility) offers a microscopic, multimodal platform that, combined with OpenStreetMap (OSM) data, enables high-resolution modeling of traffic behavior and emissions [4]. In parallel, machine learning (ML) techniques allow for the extraction of patterns from large datasets offering predictive capabilities that surpass traditional statistical methods [5]. However, most similar studies rely on statistics, satellite imagery, or vehicle characteristic data without fully considering the area layout, which limits their applicability in real-world scenarios [6]. In contrast, this study uses OSM maps, which provide an accurate and faithful representation of the analyzed area.

In the context of Quito, previous studies have demonstrated the availability of data and have considered the variability of traffic patterns throughout the day. Some approaches employ specific routes and methodologies that integrate travel time and local characteristics to estimate vehicular CO_2 emissions [7]. However, these analyses are often constrained by limited route coverage and data availability, which pose significant challenges for achieving realistic simulations that accurately reflect geographic and traffic dynamics. To address these limitations, this research proposes a framework that leverages extensive SUMO simulations on a detailed map of the Historic Center of Quito, combined with machine learning models to efficiently predict CO_2 emissions. By doing so, it minimizes the need for repeated, computationally expensive simulations while maintaining high predictive accuracy. The primary objective of this study is to develop a machine learning model capable of forecasting CO_2 emissions based on key traffic parameters such as vehicle density and fleet composition. This model is trained on a comprehensive dataset generated through realistic SUMO simulations. Furthermore, it is deployed in a web application built with Streamlit, enabling decision-makers and urban planners to easily explore different traffic scenarios and estimate emissions in real time. This approach not only facilitates the planning of sustainable mobility strategies in Quito but is also adaptable to other high-altitude urban areas facing similar environmental challenges.

2 State of the Art

Recent studies have demonstrated the potential of deep learning models such as LSTM, BiLSTM, CNN, and DCNN for emission prediction, achieving high accuracy using metrics like MSE, RMSE, and R^2 (Al Nefaie and Aldhyani, 2023; Isaac et al., 2024). However, these approaches rely on generic Kaggle datasets that lack geographic context, real traffic dynamics, or urban simulation, limiting their applicability to high-altitude environments. Singh and Dubey (2023)

addressed this by incorporating OBD-II sensor data to capture real-world driving behavior, though their method faces scalability challenges in dense urban settings. Roy et al. (2014) achieved high precision ($MAPE < 1.7\%$) using ANN models trained on experimental engine data, but their findings are restricted to controlled environments and specific engine types.

In Ecuador, Llanes Cedeño et al. (2018) conducted high-altitude vehicle emission measurements using different fuels but without predictive modeling. Moyano Tobar (2017) estimated CO_2 and NO_x emissions through microsimulation in Cuenca, although the study lacked fleet composition details and broader traffic estimates. Mena et al. (2018) simulated vehicle flow in Quito using SUMO and OSM data, contributing to mobility analysis but without including emission modeling. Finally, Wang et al. (2024) used Random Forest with driver behavior variables (e.g., harsh braking, speeding), achieving high accuracy ($R^2 = 0.975$), but their model depends on specialized sensors, reducing scalability. Overall, despite valuable contributions, most studies present limitations such as lack of adaptation to high-altitude urban contexts, reliance on generic or controlled datasets, or technical barriers for large-scale implementation. To address these gaps, this research proposes a simulation-based, context-aware methodology tailored to Quito—a densely populated, high-altitude city—capable of predicting CO_2 emissions using interpretable and scalable inputs.

3 Methodology

The present study for the analysis of carbon dioxide emissions was divided into three main stages which are: calibration and simulation, dataset construction and the production of the machine learning model. As can be seen in Fig.1, each one of these phases was meticulously analyzed in order to develop a reliable predictive model for urban areas with high traffic such as the historical center of Quito, Ecuador, which is the study area of this research. Among the tools available for traffic simulation, there are multiple options such as Aimsun, PTV VISSIM, and others. However, these are paid software, unlike SUMO, which is open-source and widely accepted by the research community.

3.1 Calibration and Simulation

Before running the simulations, a calibration phase was performed to define key parameters and ensure realistic results. The map of Quito’s historic center one of the city’s most congested areas was obtained and cleaned using OpenStreetMap with Python and netconvert. Then, netedit was used to fix conversion errors, remove unnecessary peripheral roads, adjust intersections, and complete missing elements, ensuring an accurate road network. To configure vehicular traffic, SUMO allows for a customized setup of vehicles and their distribution. For this reason, the vehicle fleet was defined and calibrated. The real distribution of registered vehicles in Ecuador, based on INEC data [16], was used as a reference, where vehicles are categorized as motorcycles, automobiles, SUVs, pickups, and

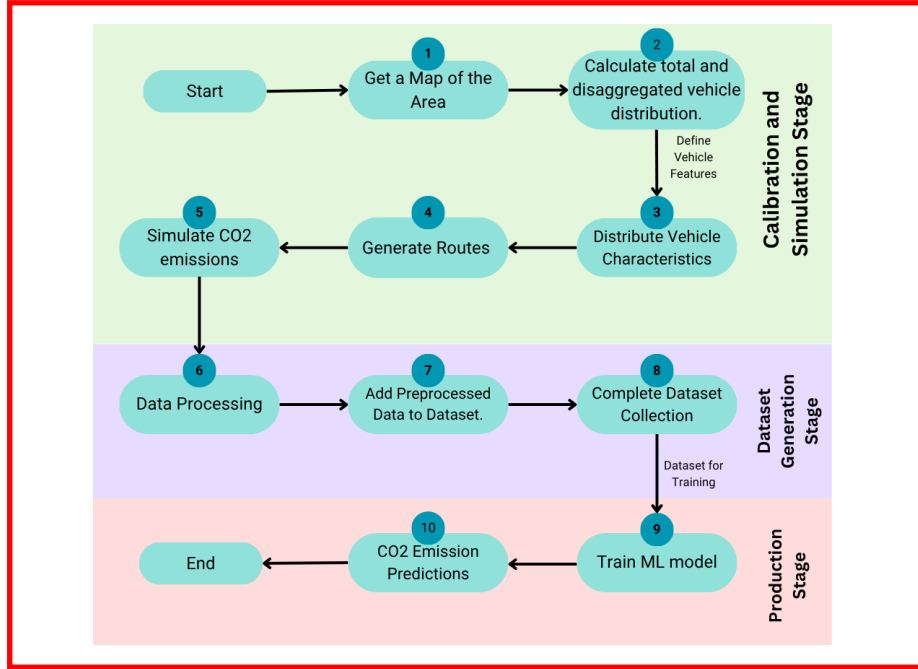


Fig. 1. Diagram of the methodology used in the investigation

trucks. Additionally, in the simulations carried out, variation ranges within an acceptable margin were applied to introduce slight differences between simulations and make the dataset more robust. The obtained ranges are shown in Table 1.

Table 1. Vehicle type distribution and variation ranges used in the simulation.

Vehicle Type	Registered Percentage (%)	Variation Range (%)
Motorcycles	29.42	26.15 – 33.95
Automobiles	28.93	24.50 – 32.73
SUVs	19.87	16.39 – 22.90
Pickups	17.32	14.36 – 20.66
Trucks	4.46	1.67 – 7.62

In SUMO, vehicles can be configured using various physical and behavioral parameters, allowing for accurate simulation of real-world traffic conditions. For this study, the most sold vehicles in 2023 were taken as a reference, based on data from AEADE [25], in the case of motorcycles, a standard measurement was taken since there is no specific model as the best seller, but rather a category. From which several configuration parameters were obtained. Additionally, the Ley Orgánica de Transporte Terrestre, Tránsito y Seguridad Vial del Ecuador was considered, and standard values were used for other parameters. In Table 2

you can see the configuration that was used for vehicle class where later it will be explained how each one of them was configured.

Table 2. Parameters used in the simulation for each type of vehicle based on the different models.

Parameters	Motorcycle	Automobile	SUV	Pickup	Truck
Base Model	Generic[28]	Kia Soluto[26]	Kia Sonet[30]	Chevrolet D-Max[27]	Chevrolet NLR[29]
Acceleration (m/s^2)	4.98	1.76	1.73	1.59	0.66
Deceleration (m/s^2)	4.5	4.5	4.5	4.5	2.0
Max Speed (m/s)	13.89	13.89	13.89	13.89	11.11
Speed Factor	1.2	1.2	1.2	1.2	1.2
Speed Deviation	0.2	0.2	0.2	0.2	0.2
Length (m)	1.5	4.3	3.9	5.295	4.76
Width (m)	0.85	1.7	1.8	1.86	1.81
Height (m)	1.35	1.46	1.64	1.79	2.18
Sigma	0.8	0.8	0.8	0.8	0.8
Tau (s)	0.57	0.57	0.57	0.57	0.75
Min Gap (m)	3.0	3.0	3.0	3.0	3.0
Lat. Alignment	center	center	center	center	center
Min Lateral Gap (m)	1.5	1.5	1.5	1.5	1.5
Car Follow Model	IDM	IDM	IDM	IDM	IDM
Emission Class	MC_4S_Euro-4	PC_G_EU4	LDV_G_EU4	LCV_Diesel_Euro-4	HDV_D_EU4

Physical Parameters: SUMO allows the definition of vehicle length, width, and height, adjusted based on manufacturer specifications.

Acceleration and Braking Parameters:

- **Accel:** The acceleration parameter is how long it takes a vehicle to get from 0 to $100 km/h$ in this parameter was taken into consideration the official data of the manufacturers and according to studies every 1000 meters of altitude the vehicle engine loses 10% of its power in the case of Quito its altitude is 2850m which means that it loses 28.50% this is reflected in the following calculation [17]:

$$P = 1 - \frac{28.5}{100} = 0.715 \quad (1)$$

$$100 km/h = 27.78 m/s \quad (2)$$

$$a = \frac{v_f - v_i}{t} = \frac{27.78}{t} \quad (3)$$

$$a_Q = a \cdot P = a \cdot 0.715 \quad (4)$$

- **Decel:** Represents safe braking without losing control. For light vehicles, typical values are between $3.0-4.5 m/s^2$ [18], and $2-3 m/s^2$ for heavy vehicles [19].

Driver Behavior Parameters: The parameters related to driver behavior included a sigma value of 0.8, which represents the randomness in driving and reflects average driver variability [20], and a tau (reaction time) of 0.57 seconds,

which studies identify as a standard value for drivers aged between 29 and 39 years [21].

Speed Parameters:

- **MaxSpeed:** Urban limits in Ecuador are 50 km/h (13.89 m/s) for light vehicles and 40 km/h (11.11 m/s) for heavy vehicles [22].
- **SpeedFactor:** It allows a moderate speeding according to the present law, values of 60 km/h and 50 km/h for light and heavy vehicles, respectively, are considered [22].
- **SpeedDev:** It is the percentage in relation to the normal speed that the driver exceeds the speed limits. In this case 20% random variability was set to reflect actual behavior.

Distance and Alignment Parameters: The parameters established by Ecuadorian traffic law include a minimum front gap (MinGap) of 3 meters, a lateral clearance (MinGapLat) of 1.5 meters to allow safe overtaking or lane changes, and a lane alignment (LatAlignment) setting where vehicles remain centered within their lanes.

Car Following Model: The IDM model was used because it is one of the models with the most realistic approach to acceleration and braking, essential for an accurate simulation of emissions. It uses the equation [23]:

$$a = a_{\max} \left[1 - \left(\frac{v}{v_0} \right)^\delta - \left(\frac{s^*(v, \Delta v)}{s} \right)^2 \right] \quad (5)$$

$$s^*(v, \Delta v) = s_0 + vT + \frac{v\Delta v}{2\sqrt{a_{\max}b}} \quad (6)$$

Here, a represents the vehicle acceleration, v is the current speed, v_0 is the desired speed, and δ is an exponent typically set to 4. The term s denotes the gap to the vehicle ahead, Δv is the speed difference relative to that front vehicle, s_0 is the minimum safe distance to maintain, T is the driver's reaction time, and b indicates the comfortable deceleration.

Emission Class Model: We used the emission models of each category corresponding to a Euro4 emission which, according to Ecuadorian law, vehicles must comply with at least to circulate in the country [24].

Finally, different vehicle densities (from 50 to 275 vehicles/km^2) were used to simulate various levels of congestion. The routes were generated with the randomTrips tool, configuring each simulation so that vehicles circulated from the first second and remained active for the entire hour. An average of 7 minimum nodes per trip was set in order to extend the routes and a 35% dynamic entry/exit probability (fringe factor) was applied to simulate vehicles entering and exiting the map frequently, which allowed creating a more realistic urban traffic. In addition, a random seed was used in each run to avoid repetition and ensure diverse results. Although this study focuses on vehicle density and fleet

composition, additional variables such as traffic signals and road slopes are directly obtained from the map extracted from OSM, while driving habits were incorporated in SUMO using the IDM model, bringing simulations closer to a realistic representation of human behavior.

3.2 Dataset Collection

Following each simulation, SUMO generated two essential output files: one containing emissions data and another detailing vehicle trajectories. In total, 1,000 files of each type were processed to build a dataset suitable for exploratory data analysis (EDA) and machine learning model training. To streamline this workflow, an automated script was developed that converted the original .xml files to .csv format and removed the first 200 seconds of each simulation, which correspond to SUMO’s calibration phase that typically introduces instability and noise. This ensured that exactly 3,600 seconds (equivalent to one hour) of clean data per simulation were retained and organized into a structured data frame.

The trajectory files (SumoTrace) captured second-by-second details of each vehicle’s movement, including time, position, type, and coordinates. From these records, four key metrics were extracted: the effective circulation area (computed from the spatial bounds of vehicle positions), vehicle density (expressed in vehicles per km^2), total distance traveled by each vehicle type (derived from accumulated position differences), and fleet composition (percentage distribution of each vehicle type within the simulation). For emissions, the analysis focused exclusively on CO_2 , disregarding other pollutants. To reduce short-term fluctuations and enhance dataset stability, an Exponential Weighted Moving Average (EWMA) was applied to the temporal emission series [31]. Total CO_2 emissions by vehicle type were estimated by averaging individual vehicle emissions and aggregating them within each category.

The final dataset comprised 1,000 structured records, each representing one hour of simulated traffic. Each record included variables such as vehicle density (vehicles/ km^2), total distance traveled, total number of vehicles, circulation area, fleet composition percentages (for Automobiles, Motorcycles, SUVs, Pick-ups, and Trucks), distance traveled by each vehicle type, CO_2 emissions by vehicle type, and total CO_2 emissions. This comprehensive dataset provided the foundation for subsequent exploratory analysis and machine learning modeling.

3.3 Model selection

After constructing the dataset, three machine learning models were trained and evaluated to predict both total and category-specific CO_2 emissions: Linear Regression, Random Forest, and an Artificial Neural Network. These models were selected to cover a spectrum of complexity: a simple interpretable model, an ensemble method, and a nonlinear approach capable of capturing intricate patterns. Linear Regression is widely used due to its straightforward interpretation of coefficients, Random Forest offers robustness and effectively handles nonlinear

relationships such as those arising from vehicle distribution, and the neural network is particularly suited for uncovering complex dependencies despite being computationally more demanding. The dataset was divided into 80% for training and 20% for validation. Additionally, for the neural network, 10% of the training set was reserved for internal validation to monitor learning and prevent overfitting.

- **Linear Regression:** Selected for its simplicity and the direct interpretability of how input variables influence the outputs.
- **Random Forest:** Configured with 100 decision trees to balance computational efficiency with predictive power, offering strong performance in capturing nonlinear interactions.
- **Artificial Neural Network:** Implemented with three hidden layers of 128 neurons each and ReLU activation functions, designed to explore deeper, complex relationships within the data.

To evaluate model performance, three common regression metrics were used. Root Mean Squared Error (RMSE) measures the average prediction error, placing greater weight on larger deviations. The Coefficient of Determination (R^2) indicates the proportion of variance in the target variable that is explained by the model. Mean Absolute Percentage Error (MAPE) expresses prediction accuracy as a percentage, making it easier to compare across different scales and scenarios.

4 Results and discussion

Before starting the dataset analysis, the results from a single simulation were reviewed to understand the general behavior of the system. In Fig.2, we can observe the temporal evolution of the average data obtained per vehicle type. It is important to highlight the need to remove the first 200 seconds of data as they correspond to the SUMO calibration period. Similarly, the application of EWMA smoothing plays a key role in minimizing the presence of outliers and improving dataset quality.

In Fig.3, the total vehicle density over time is shown, where we can see that the densities remain relatively constant—this indicates that the simulation is functioning properly. The main results of this simulation are summarized in Table 3, where it becomes clear that automobiles are the main polluters, followed by pickups, SUVs, trucks, and motorcycles, respectively. This behavior is mainly due to the fact that larger engines tend to emit more gases. While trucks typically have the largest engines and could pollute the most, their very low vehicle concentration in the simulation causes their total emissions to be surpassed by automobiles, which appear more frequently. Although motorcycles also have a high presence, their emissions are minimal due to their smaller engine size and therefore go almost unnoticed.

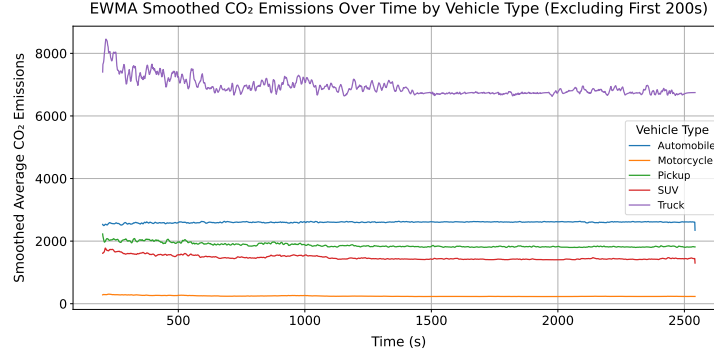


Fig. 2. Evolution over time of the average carbon dioxide produced by the different vehicles in a simulation with data processing.

Table 3. Summary of results obtained in a simulation with a vehicle density of 200 vehicles.

Vehicle Type	Distance Traveled (km)	Vehicular Distribution (%)	CO ₂ Emitted (tons)	CO ₂ Emission (%)
Automobile	411.09	29.58	0.876	43.64
Motorcycle	349.43	29.21	0.077	3.85
SUV	240.24	19.86	0.327	16.27
Pickup	241.73	16.66	0.357	17.77
Truck	54.40	4.69	0.371	18.47

4.1 Exploratory Data Analysis

The final dataset, obtained after processing, contains 1000 samples and 20 descriptive variables based on the simulations conducted using the historic center map, as explained in the methodology. The predictive model developed is designed to estimate six output values: the total CO_2 emissions and the disaggregated emissions by vehicle type.

In the Fig.4 displays the correlation between all variables. A strong collinearity is evident between the number of vehicles and vehicle density, both directly related to emission levels. In fact, when looking at vehicle density with respect to carbon dioxide emissions, we can see a correlation close to 1, which means that the problem is linear, which facilitates the training process since it is not necessary to use more complex models. In addition, the distance traveled shows a negative correlation with emissions, which can be explained by the fact that the less congestion the greater the distance traveled. Conversely, more congestion causes vehicles to idle for longer periods of time, which can increase emissions despite traveling a shorter distance. For this study, the input variables selected are vehicle density and the percentage of each vehicle type, as these are values that users can easily estimate to predict carbon dioxide emissions.

In the dataset, the vehicle composition maintains a high percentage in line with the figures reported by INEC 2023, as mentioned earlier. However, there are slight variations, as during the simulation process SUMO slightly adjusted these

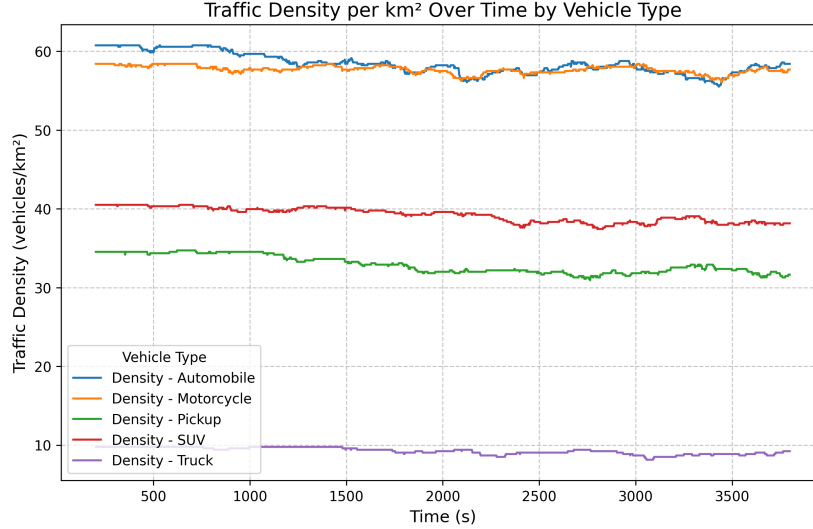


Fig. 3. Evolution of the total vehicle density within the simulation separated by the different categories of vehicles.

proportions for each vehicle type, generating small differences in the final dataset and increasing its variability for training. It is important to note that the data used are simulated using SUMO and have been validated by comparison with measurements provided by the Environmental Protection Agency (EPA), so the results should be interpreted as estimates based on modeled conditions.

4.2 Training with machine learning models

To evaluate the performance of different machine learning models in predicting CO_2 emissions, the total vehicle density along with the percentage of each vehicle type was used as input, allowing to better capture the composition of the simulated fleet. This configuration reflects variables that can be easily estimated by users, maintaining the practical focus of the study. The results obtained, summarized in Table 4, show that the linear regression model achieves good performance thanks to the existing linear relationship between density and emissions, achieving an R^2 of 0.9807. However, the neural network model showed low performance mainly due to the limited quantity and diversity of available data, which restricted its generalization capability. Strategies such as collecting additional data, augmenting the dataset with synthetic data generation techniques, and more exhaustive hyperparameter optimization could significantly improve its performance in future studies.

Tests show that changes in vehicle density generate proportional variations in the estimated emissions, reflecting the linearity present in the data. Similarly, variations in fleet composition percentages produce proportional changes in both

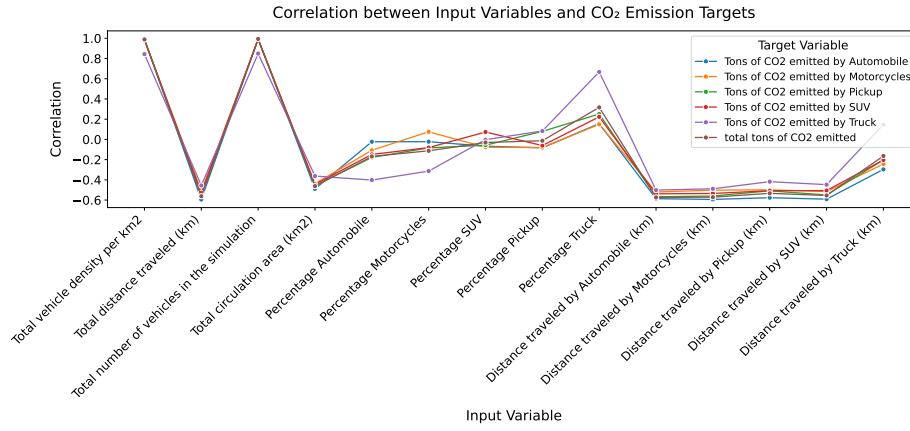


Fig. 4. Correlation of the different variables of the dataset with the target variables for prediction.

type-specific and total emissions, as well as proportional decreases depending on the redistribution of these percentages. On the other hand, the Random Forest model takes better advantage of the information on vehicle percentages, achieving the lowest error (RMSE of 0.0206), the highest explanatory capacity (R^2 of 0.9875) and the lowest percentage error (MAPE of 3.61%), positioning itself as the most accurate and robust model of the study.

Table 4. Performance obtained from the training with different machine learning models.

Model	RMSE	R^2	MAPE (%)
Linear Regression	0.0270	0.9807	5.16
Random Forest	0.0206	0.9875	3.61
Neural Network	0.0380	0.9061	8.38

In the Fig.5 presents a scatter plot comparing actual and predicted CO_2 emissions using the Random Forest model. The closeness of the points to the diagonal line reflects the model's high accuracy, supported by its low RMSE and high R^2 . This result shows the model's ability to capture both the linear relationship with vehicle density and the more subtle, nonlinear effects of fleet composition. Unlike other models, Random Forest requires no feature scaling or linear assumptions and naturally handles interactions between variables, making it particularly suitable for heterogeneous traffic simulation data. It is important to highlight that this study leveraged a high-performance computing server equipped with 128 processing cores and 1 TB of RAM for efficiently executing the extensive vehicular simulations and processing the resulting large-scale data.

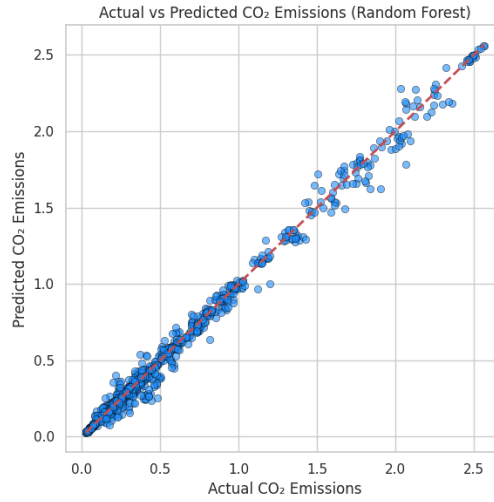


Fig. 5. Scatterplot of predicted and actual results of the random forest model

4.3 Web Application with Streamlit

After training and exporting the machine learning model, it was essential to provide an interface that makes predictions accessible, particularly for non-technical users. The core purpose of this study was not only to perform multiple SUMO simulations to build a robust dataset and train an accurate predictive model, but also to deploy this model within an interactive platform. The platform follows a simple and intuitive workflow: users first access a clean, uncluttered interface where they are guided to enter the required traffic parameters. Step-by-step prompts and clear labels ensure that even non-technical users can navigate the process without difficulty. Once the data is entered, the system processes the information instantly and transitions to the results view. Unlike tools designed for data scientists or simulation experts, this application targets decision-makers, public agencies, urban planners, and citizens interested in understanding the environmental impact of vehicular traffic. Its aim is to promote awareness, support the development of mobility policies and sustainability strategies, and offer simplified technical insights for urban management.

While frameworks such as Flask, FastAPI, or large-scale cloud services exist for deploying machine learning models, they often involve more complexity than needed for the straightforward functionality required in this study. Streamlit was therefore selected as the most appropriate choice, given its focus on data science workflows and its capability to transform Python scripts into interactive web applications with minimal overhead. The interface allows users to configure urban traffic scenarios by inputting two primary parameters: total vehicle density (in vehicles per square kilometer) and the percentage composition of the fleet (motorcycles, cars, SUVs, pickups, and trucks), as illustrated in Fig. 6. Upon

submission, the application predicts both total CO_2 emissions and emissions broken down by vehicle type. Input fields are organized logically, with numerical boxes that prevent format errors. Real-time feedback and validation ensure that users can adjust parameters quickly and see the effect of changes without needing to reload the page. This design minimizes the learning curve and enhances accessibility.

Beyond displaying numerical outputs (Fig. 7) this results view is the final step in the workflow, presenting the predicted emissions in both numerical and graphical formats. Users can seamlessly switch between different visualizations or proceed directly to exporting their results. The application includes interactive visualizations to aid interpretation. Fig. 8 shows a bar chart detailing emissions by vehicle type, while Fig. 9 presents a donut chart that illustrates the proportional contribution of each category to total emissions. As an added feature, results can be exported in various formats, including CSV, Excel (with embedded charts), and comprehensive PDF reports. This versatility makes the application a valuable tool for supporting technical analyses, strategic planning, and decision-making processes focused on sustainable urban development.

The screenshot shows a configuration interface with several input fields and a 'Predict emissions' button. The fields are organized into two columns. The left column contains: 'Vehicle density (veh/km²)' with a value of 150, '% Motorcycles (26.15 - 33.95)' with a value of 29.42, '% SUVs (16.39 - 22.90)' with a value of 19.87, and '% Trucks (1.67 - 7.62)' with a value of 4.46. The right column contains: '% Automobiles (24.50 - 32.73)' with a value of 28.93, '% Pickups (14.36 - 20.66)' with a value of 17.32, and a 'Total percentage sum: 100.00%' label. A 'Predict emissions' button is located at the bottom left.

Fig. 6. Configuration of vehicle density and distribution parameters.

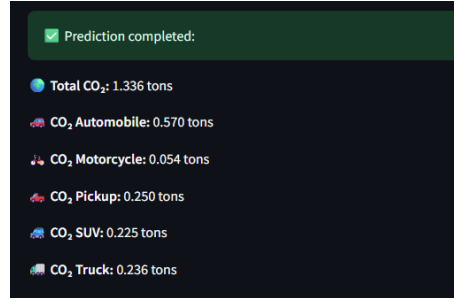


Fig. 7. Numerical results of CO_2 emission predictions.

The model can be easily updated with new simulation data or real measurements. It can be retrained with additional information, maintaining its accuracy and adaptability to changes in fleet composition or traffic conditions.

5 Conclusion

This study demonstrated the effectiveness of integrating detailed traffic simulations with machine learning techniques to predict CO_2 emissions in urban environments, specifically illustrated through the case of Quito's Historic Center. By leveraging SUMO and OpenStreetMap data, a comprehensive dataset comprising 1,000 independent simulations was created, capturing diverse traffic densities and fleet compositions. This dataset enabled the training and compar-

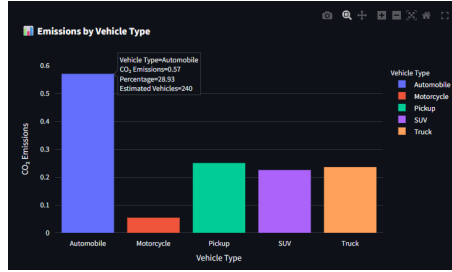


Fig. 8. Interactive bar chart of predicted CO_2 emissions by vehicle type.

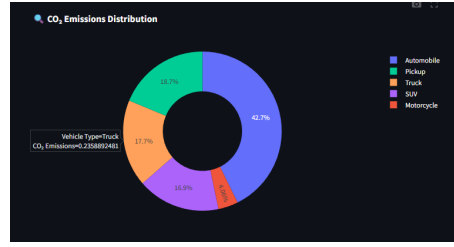


Fig. 9. Donut chart illustrating the percentage contribution to total CO_2 emissions.

ison of multiple predictive models, with the Random Forest model achieving the highest accuracy.

The primary outcome of this research is a machine learning model capable of estimating both total and category-specific CO_2 emissions in just seconds, using only two input parameters: overall vehicle density and the percentage composition of the fleet by vehicle type. This significantly reduces the computational burden of running extensive new SUMO simulations for each scenario. The model serves as a practical decision-support tool for policymakers, urban planners, and environmental agencies, allowing them to rapidly explore the impact of traffic configurations. By adjusting inputs such as fleet composition—favoring cleaner vehicles—or simulating reduced congestion, users can assess potential emission reductions and formulate targeted strategies. This provides a robust quantitative foundation for awareness campaigns, regulatory measures, and climate action initiatives aimed at sustainable urban mobility. Furthermore, deploying the model through an interactive Streamlit web application makes these predictions highly accessible. Non-technical users can intuitively explore scenarios, generate visual reports, and make informed decisions backed by data.

The proposed methodology is based on the best-selling vehicle models in Ecuador, so applying it in another country should take this reference into account for future calibrations. In high-altitude cities, it could be applied by changing the simulation map or using the obtained model as a reference for areas with geography similar to Quito. Although this work focuses on CO_2 , the methodology can be extended to other relevant pollutants such as NOx or PM_{2.5}, enabling a more comprehensive assessment of urban air quality.

Future work will focus on extending this methodology to other urban areas, experimenting with higher traffic densities and broader fleet variations, and incorporating additional pollutants like NOx and PM_{2.5} to enable a more comprehensive environmental evaluation.

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